Semester-Long Project Proposal

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1. **Project Overview:**
   This project applies machine learning to NBA data to tackle two linked prediction tasks:
   a. Classification - predict the probability that a given team wins a game (win/loss outcome).
   b. Regression - predict individual player point totals for a given game.
   Both tasks draw on the same feature-engineering foundation built from historical box scores, letting us compare how classification and regression approaches perform on related sports-analytics problems.

2. **Background information:**
   Sports outcome prediction is a well-known problem: even strong models rarely exceed 65-70% accuracy on NBA game outcomes, and betting markets set a high baseline. This gives us a realistic, meaningful target and forces genuine feature engineering rather than model-fitting. The project will produce insights (which factors most influence winning, how player output varies with context) alongside predictive models.

3. **Datasets:**
   Primary: *Kaggle "NBA Dataset: Box Scores and Stats (1947–Today)" (Eoin Moore)*: Comprehensive historical box scores for all games since 1947, with advanced stats from 1997 onward, updated nightly. This single source provides game results, team box scores, player box scores, and advanced metrics, which is the backbone for both the game-outcome (classification) and player-performance (regression) tasks. I'll use about 5 recent seasons for a manageable, meaningful sample.
   Supplementary: *Basketball Reference*: Used to cross-check data quality and to back up the feature set with any derived/advanced metrics not present in the primary dataset (for example, specific opponent-adjusted efficiency measures). Given the Kaggle dataset's range, Basketball Reference primarily serves as a validation and backup source.
   Integration approach: I'll join records using shared keys (game ID, team, date, player ID). I'll fix any name or ID mismatches between sources, remove duplicates, and check that the numbers agree across sources.
   Why these sources: The Kaggle dataset offers broad historical coverage and combines game, player, advanced data in one place, avoiding live-API rate limits and setup overhead. Basketball Reference adds backup and cross-validation. Together, they give a good foundation for feature engineering.

4. **Method:**
   a. Collect and clean data

b. Build features
c. Build models
    i. Classification
    ii. Regression
d. Train/Test model
e. Analyze

5. **Features:**
    a. Recent team form (last-5 and last-10 game averages for points, ratings, etc.)
    b. Rest days, back-to-back games, home or away
    c. Winning/losing streaks and past matchups between the two teams
    d. Opponent strength / how hard the schedule has been
    e. For the player model: the player's recent form, minutes trend, and how strong the opponent's defense is
6. **Timeline:**
    - Week 1: Collect data, set up tools, explore the data; load Kaggle dataset and cross-check with Basketball Reference
    - Week 2: Clean data; build game-level and player-level tables; build features (recent form, rest, home/away, opponent strength)
    - Week 3: Build and compare models; baselines (logistic + linear regression) then stronger models (XGBoost / random forest); set up time-based testing and tune settings
    - Week 4: Evaluate results, check feature importance, study errors; make charts; write report and prepare presentation